A Hybrid Design Automation Tool for SAR ADCs in IoT

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***Abstract*— In this paper, a hybrid design automation tool for asynchronous successive approximation register analog-to-digital converters (SAR ADCs) in Internet-of-Things applications is presented. The circuit design-driven tool uses a top-down design approach and generates circuits from specification to layout automatically. A hybrid approach is introduced for different circuits of a SAR ADC: fully synthesized control logic; a script-based flow combining equations, library, and template-based design for the digital-to-analog converter; a lookup table approach combined with selective simulation-based fine tuning and template-based layout generation for the sample and hold; library-based comparator design and script-based layout generation. By balancing the automation and manual effort, the circuit design time is reduced from days down to minutes while still being able to maintain ADC performance. The proposed flow generated two ADC prototypes in 40-nm CMOS, an 8-bit 32 MS/s and a 12-bit 1 MS/s SAR ADC, and enabled excellent power efficiency. The two ADCs consume 187 and 16.7 μW at 1-V supply voltage, achieving 30.7 and 18.1 fJ/conversion-step, respectively.**

***Index Terms*— Design automation, hybrid approach, low power, successive approximation register analog-to-digital converter (SAR ADC).**

I. Introduction

TODAY, many system-on-chips (SoCs) in Internet-of-Things (IoT) applications consist of both digital and analog circuitry for a high integration level and low cost. The highly integrated chips usually consist of millions or billions of devices, and this results in a huge circuit design effort. The digital circuit design has already been automated by many mature commercialized tools and flows, from behavior description to layout generation. The analog circuits are often manually designed for both schematic and layout. As a result, although the analog circuit scale in an SoC is usually small compared to that of the digital part, the design time for analog circuits is still long. While more and more functions are moved from analog domain to digital domain, analog-to-digital converters (ADCs) as an interface between analog domain and digital domain are still required. Many wireless sensor nodes (WSNs) in IoT/Internet-of-Everything applications (e.g., Bluetooth low energy and IEEE 802.15.4) require low-power ADCs with varying resolution (8∼12 bit) and speed (∼MS/s). To achieve an optimum performance, each ADC has to be customized, increasing the design cost and time. To overcome this issue, circuit design automation is desired for ADCs to minimize the design cost while still maintaining the performance.

TABLE I
SUMMARY OF THE AUTOMATION METHODS FOR FRONT-END SYNTHESIS

	*Library based	Knowledge based	Equation based	Simulation based	Artificial intelligence algorithms
Pros	Immediately ready	Fast	Fast	Accurate	General
Cons	Large preparation effort	Lack of generality	Lack of accuracy	Long simulation time	Need many training examples

*A simulation-based method is used during library construction

The design automation for analog and mixed-signal circuits includes roughly two steps: front-end synthesis and back-end generation. Front-end synthesis can include architecture selection, translation of functional specification to subcircuit specification, and device sizing, and back-end generation can include layout generation of sub-blocks as well as floor planning and routing.

The approaches for the front-end synthesis can be roughly divided into five categories as shown in Table I: library-based approach, knowledge-based approach, equation-based approach, simulation-based approach, and artificial-intelligence-based approach. A library-based approach requires a large effort to create a new library for each block in each technology. This may be acceptable for a single simple cell but not for a complicated system, which includes many blocks. The knowledge-based approach encodes specific heuristic design knowledge from experts into a design plan that is used during the synthesis of the analog circuit [1]. Specification inputs will be translated to the topology selection and the unique solution of the circuit sizing following the

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design plan. However, a content-independent design plan is difficult to make, which limits the use of this approach. The equation-based approach uses a simplified analytic equation to formulate the performance of the circuit [2], [3]. Constrained optimization algorithms instead of a specific design plan are performed using these equations for the optimization of the circuit. Although this approach is more general, the accuracy of the result is a big problem, especially in an advanced process because the design equation has to be derived and simplified so that the optimization algorithm can be executed. To obtain higher accuracy, a full-SPICE simulation-based approach is introduced in the optimization loop [4]. The main problem with this approach is the long run time, especially if the initial search space is not well defined. In [5], an artificial-intelligence-based approach uses the idea of evolution to automate the synthesis of analog circuits. However, large amounts of well-designed examples are required for training the neural network.

For the back-end automation, four methods have been used: library-based method, script-based method, template-based method, and standard digital flow. In the library-based method, the layouts of the required components are manually designed beforehand and put together in a library, which can be directly reused later on. The layout can be precisely optimized but it takes a long design time. The script-based method [6] describes the transistors' location and interconnection in pure ASCII instead of manual place and route. The template-based method uses a template for a design and can maintain the layout shape irrespective of different sizes [7]. The standard digital flow is very mature for digital circuits' place and route but is used less often for a customized analog layout [8].

Prior-art design automation is focusing on either behavior-level, front-end synthesis or back-end synthesis and elaborates on relatively simple analog cells [1]–[5], [7], [9]–[12]. In the context of successive approximation register (SAR) ADCs, Brenna *et al.* [11] proposed a MATLAB tool that allows statistical simulation of capacitor mismatches and parasitics in the digital-to-analog converter (DAC), and a behavioral SAR ADC model is proposed in [12] which allows fast simulation. However, while these tools support with the sizing of the components and by reducing simulation time, they do not output a finalized schematic or layout of an entire SAR ADC, which is the purpose of this paper. The design automation for complete ADCs is more cumbersome due to its complicated architecture, larger scale circuitry, and more sophisticated signal processing between the analog and digital domains. Prior art tries to synthesize a whole ADC using a single synthesis flow, chosen for a specific architecture. For a stochastic flash ADC [13], the traditional analog circuitry is replaced by pure digital gates so that the whole ADC can be integrated into the digital synthesis flow. However, the resolution and performance are limited due to the stochastic flash architecture. For a $\Sigma\Delta$ ADC [8], a library-based approach and standard place and route (P&R) for layout generation have been used. Although all the analog components are still manually designed, the power efficiency is yet behind the state of the art. Wulff and Ytterdal [6] introduce a compile method for the SAR ADC but is limited to layout generation only.

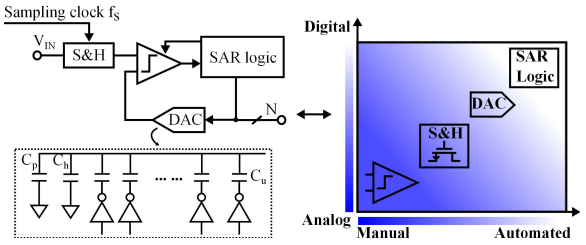


Fig. 1. Block diagram of a SAR ADC and illustration of its sub-blocks.

Huang *et al.* [14] introduce a systematic design method on a schematic level for a SAR ADC but large differences between simulation and measurement results are observed ($>10\times$).

Alternatively, instead of synthesizing all the circuits using an identical approach, a design tool based on a hybrid automation approach is introduced in this paper, allowing different automation approaches for the individual subcircuits [15]. Both the circuit and layout levels are included. With the assistance of this tool, the designer can better balance automation and customization and reduce the design time due to repetitive labor work while maintaining the ADC performance. This approach is elaborated for a SAR ADC, as the SAR ADC is popular for WSNs thanks to its excellent power efficiency [16]. The design automation approach is verified by generating two prototype ADCs. The measurements show a good agreement with the simulation results.

The remainder of this paper is organized as follows. Section II introduces the proposed hybrid automation approach applied to the SAR ADC. The specification translation of the automation approach is shown in Section III. The implementation details of the automation tool and the circuits are presented in Section IV. The measurement results are shown in Section V, and the conclusion is drawn in Section VI.

II. Hybrid Design Automation Applied to a SAR ADC

A conventional N-bit SAR ADC usually includes four main blocks as shown in Fig. 1: sample and hold (S&H), comparator, control logic, and DAC. In each SAR conversion, the S&H samples the analog input voltage and stores it on the capacitor arrays of the DAC. The DAC output will approximate the sampled input voltage in N comparisons through a successive-approximation algorithm. This is performed by the feedback loop that consists of the comparator and the SAR logic. In a synchronous SAR ADC, a high-frequency clock ($N \times f_s$) is needed but this can cause a significant overhead in power consumption. Therefore, the asynchronous operation is used in this paper [16]. Thanks to the dynamic operation of all the ADC building blocks, the power consumption is inherently scalable with the sampling frequency. Furthermore, the topologies of the S&H, comparator, and DAC are fixed to a bootstrapped sampling switch [17], a dynamic comparator [18], and a top-sampling fully binary-weighted C-DAC with a monotonic switching scheme [19], [20], as it has been shown that such topologies can already cover a large performance range in terms of speed and resolution.

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TABLE II
DESIGN TIME ANALYSIS OF A SAR ADC

Manual Design time	Front-end (inc. optimization)	Back-end (inc. optimization)	Simulation
SAR logic	days	days	minutes
DAC	hours	days	minutes
S&H	hours	hours	minutes
Comparator	days	hours	hours
ADC top	hours	days	hours

TABLE III
SUMMARY OF THE PROPOSED HYBRID AUTOMATION METHODS

	Front-end generation				Back-end generation		
	Standard digital flow	Equation based	Simulation/ knowledge based	Library based	Standard digital flow	Template based	Library based
SAR logic	√				√		
DAC		√		√		√	
S&H			√			√	
Comparator				√			√
ADC top		√	√			√	

Note that the monotonic switching scheme is not the most energy-efficient one, but thanks to the small capacitors used in the proposed DAC layout, the power consumption of the DAC is nonetheless relatively insignificant compared to that of the ADC (<10%). While the tool currently only implements the above-mentioned topologies, it could be expanded with other topologies in a similar way.

As mentioned, to ensure the performance of the ADC, a manual design is usually applied to all the four blocks, which is very time consuming (Table II). In particular, for the SAR logic, both the front-end and back-end design time are in the order of days, thus both need to reduce. For the DAC design, the design time for the back end is much more than for the front end, and therefore, reducing the back-end design time improves the total design time more. For the S&H, the front-end design and the back-end design consume a similar design time. For the comparator, most of the design time is spent in finding the proper sizes for the devices and reducing the back-end design time helps less. For the ADC top level, most of the design time is required for the back end, as a large number of analog and digital connections have to be made in the layout.

To define the best design approach, the four basic blocks of a SAR ADC are analyzed first from two aspects: analog/digital-oriented and manual/automated aspects (Fig. 1). Analog-oriented aspect implies that the circuit is physically constrained (by noise and mismatch) and expertise is needed to properly optimize the circuit, which may require manual effort. Digital-oriented aspect implies that the circuit is less physically constrained and can be abstracted and synthesized more easily without losing performance. In this paper, different approaches are adopted for the front-end and back-end designs, comprising the four basic blocks, and the top-level design of a SAR ADC as follows (Table III).

1) *SAR Logic:* Both the front end and the back end of the control logic can be synthesized using a standard synthesis flow. This has the advantage of flexibility in reconfiguring the number of bits, the operation speed, and the timing of the control signals. The design time is much shorter, up to a few hours, compared to that of the manual design, and when redesigned, only a few key parameters need to be tuned. The redesign time, including front end, back end, and simulation, can be reduced to less than 1 h. In addition, it features the possibility to integrate digital calibration logic in the future as will be discussed later. While the power efficiency of manually designed dynamic logic [16] is still better than synthesized logic, a standard synthesis flow is adopted here thanks to its advantages in design time and reusability.
2) *DAC:* The front-end design of the DAC capacitor array can be automated as follows. The DAC capacitance value is computed automatically according to the noise and matching requirements through an equation-based method as shown later. The DAC driver is simply an inverter, and therefore uses a library-based approach. Compared to the front end, the DAC back-end design usually takes much longer due to the strict requirements, e.g., symmetry and optimization of parasitics at attofarad (aF) level. Fortunately, thanks to the regularity of the DAC layout, its design can be automated through template-based programming as shown later, which reduces the majority of the DAC design time.
3) *S&H:* The S&H circuit, together with the ADC input capacitances, influences the ADC bandwidth. This effect can be easily checked by simulating the spurious-free dynamic range (SFDR). Considering the short simulation time, a simulation-based approach is introduced for the front-end design. In addition, to further optimize the S&H circuit while minimizing the manual effort, a knowledge-based approach is used by tuning only the critical devices as shown later. For the back end, a template-based layout generation through programming is used to reduce the relayout time during optimization or redesign.
4) *Comparator:* The ADC performance is very sensitive to comparator nonidealities (e.g., offset error, noise, and layout asymmetry), and therefore, the comparator design needs more attention. In the elaborated design, the comparator design is library based for both the front end and the back end.
5) *ADC Top Level:* For the front end, the ADC top-level design uses equations to translate the top-level specifications into subcircuit requirements at the beginning of the design, and a simulation-based approach to assure that the performance is met. For the back end, thanks to the regular layout geometry of a SAR ADC, a template-based layout generation is introduced, which reduces the layout time from days to minutes.

The discussed design steps are embedded in a design flow to automate a SAR ADC design in six steps (Fig. 2). The flow first takes the ADC specifications as input, then translates the input into parameters for the sub-blocks. Afterward, each sub-block is generated automatically using their corresponding method. Finally, the layout of all the sub-blocks will be assembled automatically, and the performance of the generated ADC is verified through simulations using an extracted view.

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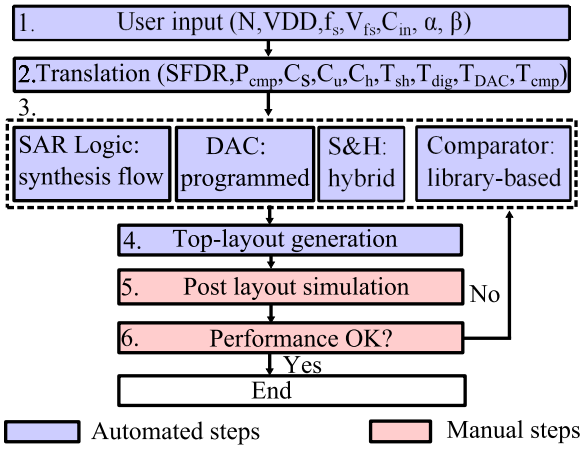


Fig. 2. Design flow of the proposed method.

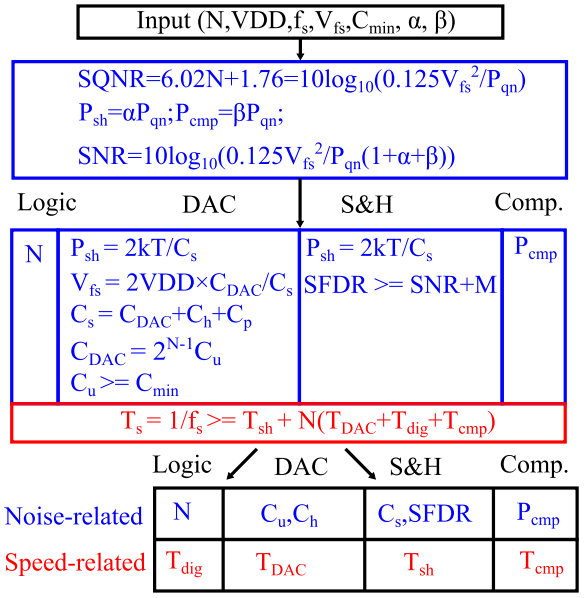


Fig. 3. Translating from ADC specifications to block parameters.

The design process will stop when the simulated ADC performance reaches the design target. Otherwise, further optimization is still possible by fine tuning the parameters of sub-blocks. In this paper, the steps 1–4 in Fig. 2 are automated while the simulation and performance checks are manual.

III. Specification Translation

The translation of the overall ADC specifications into the sub-block parameters is automated through an equation-based method, in step 2 of Fig. 2. The equations are divided into two categories: noise related and speed related, as shown in Fig. 3. The input resolution N determines the signal-to-quantization-noise ratio, which leads to the ADC quantization noise floor P_{qn} (Fig. 3), based on a provided input signal range V_{fs}. The noise requirements of the sub-blocks (the sampling noise P_{sh} and the comparator noise P_{cmp}) are computed based on this quantization noise, as shown in Fig. 3. As a result, the signal-to-noise ratio (SNR) degradation due to the extra circuit noise is $10\log_{10}(1+\alpha+\beta)$ dB. For instance, with $\alpha=\beta=(1/2)$, a 3-dB SNR degradation [equivalently 0.5-bit effective number of bits (ENOB) loss] will occur due to the noise from sampling and comparator but other choices are possible dependent on the user preference.

Then, the derived noise requirements are used to compute the circuit parameters for each sub-block. In particular, the total input capacitance C_s of the ADC can be computed from the sampling noise P_{sh} requirement. At the same time, according to the equations to calculate the input signal range of the ADC ($2\text{VDD}\cdot C_{\text{DAC}}/C_s$), the DAC capacitance C_{DAC} and the attenuation capacitance C_h are computed. From this, assuming a monotonic DAC switching scheme, the minimum unit capacitance C_u of the DAC can be computed based on the noise constraint. The mismatch constraint of the capacitors is not considered as it is expected to be tackled by mismatch calibration as in [21] as discussed later. For the S&H, the required SFDR should be M dB higher than the targeted SNR, where M is typically set to 10 dB but can be overruled by the user. For the comparator, the calculated noise requirement P_{cmp} will decide which comparator is selected from the library.

Meanwhile, the operation speed of the comparator, DAC, and logic is related to the overall ADC operation speed f_s (Fig. 3). The total time of the S&H, comparator, DAC, and logic should be no more than the period of the sampling clock T_s. By default, half of the period is used for the S&H to track the analog input signal, and the other half is used for the conversion, which consists of $N\times$ operation of the comparator, DAC, and SAR logic. In this paper, the logic and DAC are first designed and simulated, and the remaining time is for the comparator.

IV. Block Implementations and Top-Level Layout

As discussed, the requirements of the sub-blocks are derived from the equations in Fig. 3 and will be used for the design of each sub-block, respectively, and for the top-level layout generation.

A. Fully Synthesized Control Logic

The digital control logic of the asynchronous SAR ADC performs the binary search algorithm and also generates an internal clock for the asynchronous ADC. In particular, a monotonic switching scheme [19] is implemented to improve the power efficiency of the ADC. Thanks to the well-developed digital synthesis flow, most of the design effort is now taken over by the digital design tools. For example, when redesigning for a different speed, the designer only has to change the timing constraint parameter of the synthesis flow. The delay of the digital control logic will be optimized automatically by the synthesis flow, to meet the required operation speed. This is more convenient and faster compared to the iterative manual optimization method.

B. Hybrid DAC Design

The charge-redistribution DAC is often chosen for SAR ADCs thanks to its good power efficiency. It consists of DAC drivers and DAC capacitors, which are designed automatically through a script-based automation flow combining

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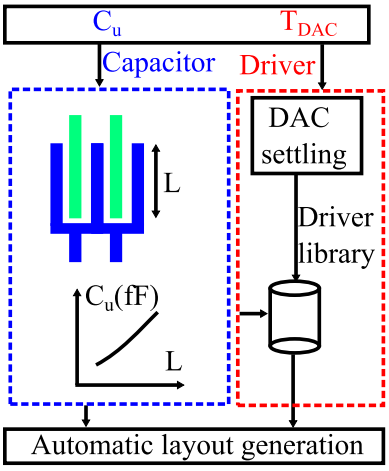


Fig. 4. DAC design automation flow.

a mix of equation-based, library-based, and template-based designs (Fig. 4). An equation-based method is first adopted to size the unit capacitor C_u based on the noise constraint as shown in Fig. 3. On the other hand, when the ADC resolution goes beyond 10 bits, C_u tends to be limited by the capacitor mismatch and thus has to be sized relatively large, degrading the power efficiency. Alternatively, to save power, C_u can be sized to just meet the kT/C noise, and calibration can be employed to compensate the capacitor mismatch errors [21]. Calibration is not yet present in the current tool but can be easily integrated into the digital circuit synthesis. In case mismatch is dominant while calibration is not used, the sizing process of C_u becomes more difficult. In that case, an assistance tool such as that in [11] could be inserted to determine appropriate sizing.

It can be calculated that for medium- and low-resolution ADCs with a rail-to-rail input range, the noise-constrained C_u is unpractically small. For example, C_u for an 8-bit ADC can be as small as 40 aF based on noise requirements, which will be much smaller than the parasitics. This is undesired since it will reduce the input signal range and make the input signal range more vulnerable to parasitics. To avoid this, the designer can optionally set a lower boundary ($C_{\min}$) for C_u, e.g., 600 aF. After selecting C_u, the DAC driver needs to guarantee that the DAC output can settle in time T_{DAC}. The selection is made from a library of drivers with various strengths, matching the selected C_u and the required operation speed f_s. The DAC driver inputs are the load of the digital control logic. When using larger size DAC drivers, the loading of the corresponding digital control bit should be adapted for the synthesis.

The DAC layout is very structured with a repetitive geometry, and therefore can be automated efficiently to save layout design time. In this paper, a programmable cell (Pcell) is used for the DAC design automation and its layout generation. By programming the key parameters of the DAC, this tool enables the users to provide the specification, automate the device sizing, and generate the layout accordingly. C_u is a customized finger capacitor as shown in Fig. 4. This allows the employment of a small unit capacitance (sub-fF) and to reduce the DAC area and power consumption at the same time. In addition, the unit capacitance can be approximated as

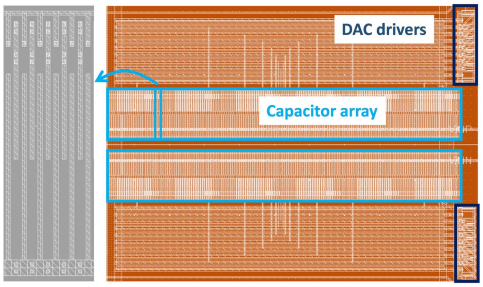


Fig. 5. Generated DAC layout example for an 8-bit ADC.

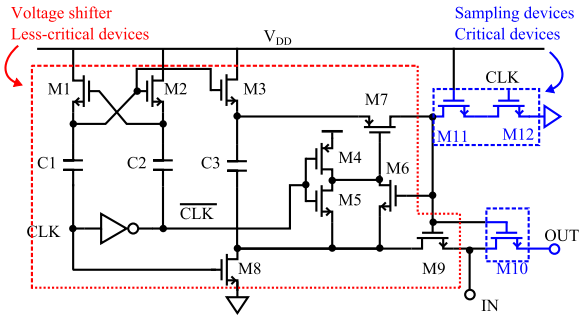


Fig. 6. S&H circuit.

linearly proportional to the overlapping length L between the top plate and bottom plate of the capacitor. Therefore, the unit capacitance can be precisely controlled by simply programming the unit capacitor length L dependent on the required value of C_u (Fig. 4). As shown in Fig. 5, the unit capacitor value C_u changes the length L, which is in the vertical direction, while the resolution N changes the total number of capacitors placed beside each other in the horizontal direction. As such, L and N can be independently modified without significantly changing the overall DAC floorplan. The DAC layout is automatically generated from the Pcell, which is described in SKILL language.

C. Hybrid S&H Design

For a given DAC capacitance, the ON-resistance R_{on} of the sampling switch determines the bandwidth of the S&H. In modern CMOS technologies, the supply voltage drops below 1 V, making it difficult to reach the desired bandwidth (∼MHz) for the ADCs in WSNs if directly controlled switches are used. Therefore, a bootstrapped circuit [17] is used in this paper, as shown in Fig. 6. As mentioned, a simulation-based approach is introduced for the S&H circuit thanks to the short simulation time for SFDR. However, noting that there are multiple devices in the S&H circuit (Fig. 6), the optimization time for the sizing would be too long to be acceptable.

To reduce the optimization time while still maintaining the good performance for the S&H circuit, a knowledge-based approach is added as follows. The circuit can be roughly divided into two parts: the devices for sampling (performance critical) and for the voltage shifter (performance less critical). A voltage multiplier (M1–M9) performs the function of boosting the gate voltage of the sampling switch M12. These devices are relatively less critical as long as the function of voltage boosting can be achieved. Therefore, the size of the

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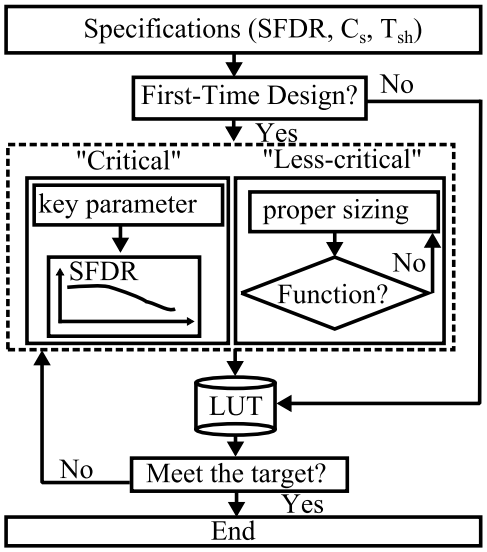


Fig. 7. S&H design automation method.

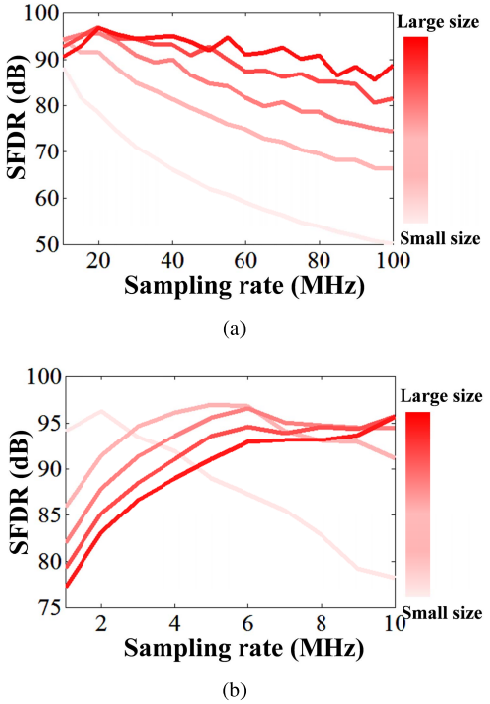


Fig. 8. Simulated SFDR of S&H with various device sizes for relatively (a) high and (b) low sampling frequency, respectively.

noncritical devices is fixed for all designs in this automation tool. M10–M12 are critical for the S&H performance, and therefore need to be carefully treated.

Therefore, a hybrid design approach, consisting of simulation- and knowledge-based approaches, is proposed for the front-end design of the S&H circuit (Fig. 7). One S&H is implemented at first (including layout) and reused as a template when redesigned for a different specification. When redesigned, the less critical devices are fixed while the critical devices are tuned for optimization. As a result, the total number of iterations needed to reach the optimum point is significantly reduced, enabling the employment of a simulation-based lookup table (LUT) approach for the S&H circuit. The LUT can be constructed based on the simulation results for various device sizes as shown in Fig. 8. In this way, in a next design, an approximately optimized device size can be directly selected from the LUT. It is crucial to choose the device dimension properly from the LUT according to the required SFDR and the ADC sampling rate. Larger devices do not always result in better SFDR but can degrade the SFDR for low-speed ADCs due to device leakage. Moreover, further optimization is still possible by further fine tuning the critical devices. To save the optimization time, one parameter, which is the width of the devices, is used for sizing the three critical devices since they are usually tuned in the same direction. Furthermore, it is worthwhile to mention that the ADC input capacitance C_s influences the SFDR of the S&H circuit. The bandwidth of the S&H is determined by an RC constant, where R is determined by the size of the sampling transistor M10 and C is C_s. This impact of C_s is accommodated indirectly by scaling R in an opposite direction to maintain a constant RC value. For the back-end design, considering that most of the devices are fixed and only three transistors are programed, a template-based layout generation through SKILL is introduced for the S&H, similarly as for the DAC, which further reduces the layout redesign time.

TABLE IV
ILLUSTRATION OF LIBRARY-BASED METHOD FOR COMPARATOR

P_{cmp} / T_{cmp}	<=P_0	(P_0, P_1]	(P_1, P_2]	(P_2, P_3]	... ...
(0, T_0]	CMP1.1	CMP1.2	CMP1.3	CMP1.4	... ...
(T_0, T_1]	CMP2.1	CMP2.2	CMP2.3	CMP2.4	... ...
(T_1, T_2]	CMP3.1	CMP3.2	CMP3.3	CMP3.4	... .
(T_2, T_3]	CMP4.1	CMP4.2	CMP4.3	CMP4.4	... .
(T_3, T_4]	CMP5.1	CMP5.2	CMP5.3	CMP5.4	... .

D. Library-Based Comparator Design

In WSNs, the comparator performance is crucial for SAR ADCs, as it can either dominate the overall ADC power consumption or degrade the ADC ENOB. Therefore, even for an experienced ADC designer, it usually takes days to optimize the speed, noise, and power efficiency of the comparator. However, it is not straightforward to correlate the comparator performance with the device sizing due to its relatively complex circuitry, nonlinear operation, and sensitivity to layout imperfections. Therefore, a library-based approach is introduced, in which comparators with different performance combinations can be selected by the computer-aided design tool when users input the specifications. The comparators in the library will have a 2-D index: input-referred noise and comparison time. The probability of metastability can be controlled by assigning a more conservative comparison time to build in some extra buffer time. According to these two requirements, a proper device in Table IV will be selected. In this paper, a dynamic comparator is used thanks to its efficiency and frequency scalability [16], [18].

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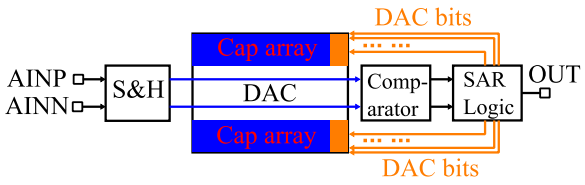


Fig. 9. Template of the ADC top layout.

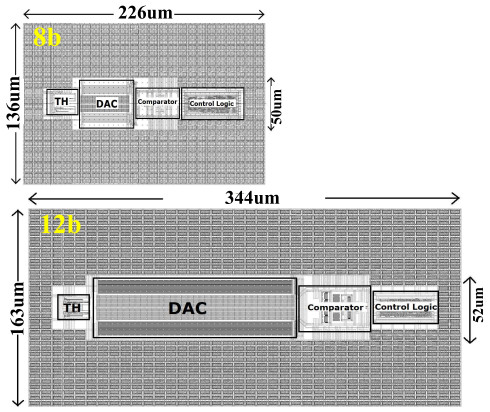


Fig. 10. Generated ADC layouts using the proposed flow.

E. Top-Layout Generation

The traditional P&R in the standard digital flow minimizes the manual layout effort for digital circuits by automatic P&R [8]. However, this can cause problems for analog circuits, which are sensitive to layout nonidealities (e.g., parasitics, asymmetries, and IR drops), and thus may degrade the ADC performance. Alternatively, a template-based layout generation is introduced [22]. This layout generation is used for well-understood designs with sufficient regularity or geometric templates. It only adapts the size for devices that are critical to optimization while the remaining devices are relatively fixed. The interconnections between the devices are adapted automatically.

The aforementioned tool for the design of the sub-blocks is developed using SKILL language and the Pcell tool in Cadence. The layout generations of the ADC top are automated using a geometrical template-based method. This method allocates each sub-block according to a template and connects the pins between each block through scripting (Fig. 9). The geometrical coordinates of all sub-blocks are parameterized, and the pin locations of each block are automatically adapted accordingly. This greatly saves the design time for layout, and as a result, it takes only a few seconds for the program to integrate into a design rule check and layout versus schematic clean compact layout (Fig. 10). Then, simulations based on an extracted view can be executed to verify the performance of the ADC. Note that it is still possible to fine tune each sub-block individually to optimize the overall ADC performance using the implemented design flow.

V. Verification With Prototypes

To verify the proposed design flow, two prototype chips with different specifications for speed and resolution have been implemented in a 40-nm CMOS: an 8-bit 32-MS/s SAR ADC and a 12-bit 1-MS/s SAR ADC (Fig. 11). The compact layout leads to a small core area of the two ADCs (Fig. 10) of 0.011 and 0.016 mm^2, respectively. Including decoupling capacitors, they occupy 0.031 and 0.056 mm^2.

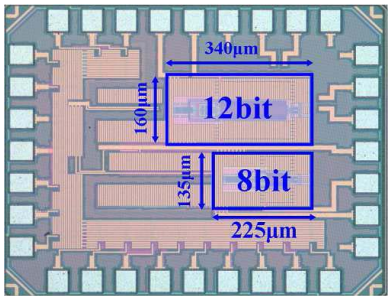


Fig. 11. Die photograph of the prototype ADCs in 40-nm CMOS.

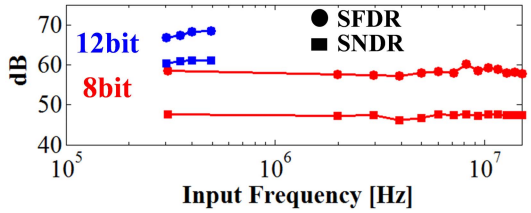


Fig. 12. Dynamic performance of the two ADCs.

The 8-bit ADC consumes 187 μW at 32 MS/s and 1-V supply and achieves 47.4-dB SNDR up to the Nyquist frequency (Fig. 12). This leads to a 30.7-fJ/conversion-step figure of merit (FoM) as defined in 1, which is comparable to the simulated 25.1 fJ/conversion-step, considering that simulation results did not include nonidealities, e.g., noise and mismatch-esMing Ding

$$\text{FoM} = \frac{\text{Power}}{2^{\text{ENOB}}\min(f_s, 2\text{ERBW})} \tag{1}$$

The 12-bit ADC consumes 61.1 μW at 1 MS/s and 1 V, which is close to the simulation results. Considering that the DAC capacitors are sized for kT/C noise instead of mismatch, large integral nonlinearity (INL)/differential nonlinearity (DNL) errors are measured, as shown in Fig. 13, thus limiting the ADC performance. However, this error is assumed to be solved by the DAC mismatch calibration [21]. The measured input-referred noise of the 12-bit ADC is 0.47 LSB_{rms}. This indicates that the noise-limited ADC SNDR is above 11 bit once the DAC mismatch-induced error is corrected.

In addition, both ADCs achieve stable SNDR/SFDR up to the Nyquist frequency (Figs. 12 and 14), indicating a sufficient performance for the S&H circuits. Overall, the measured power consumption and speed match with the simulated numbers. We should note that the simulated numbers are not estimated by the proposed tool but by postlayout simulation of the layout which is generated by the tool. The measured FoM for the 8-bit ADC is close to expectation (as shown later). For the 12-bit ADC, the FoM correspondence is worse compared to the 8-bit ADC due to the mismatch-induced errors. However, the correspondence is better than the prior art.

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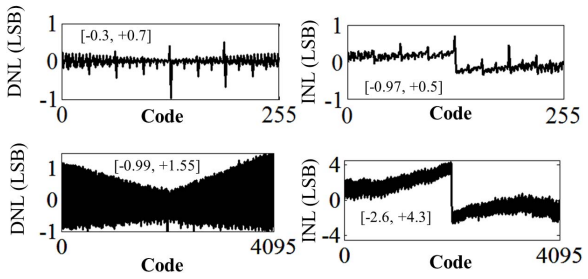


Fig. 13. Measured INL/DNL of the two ADCs.

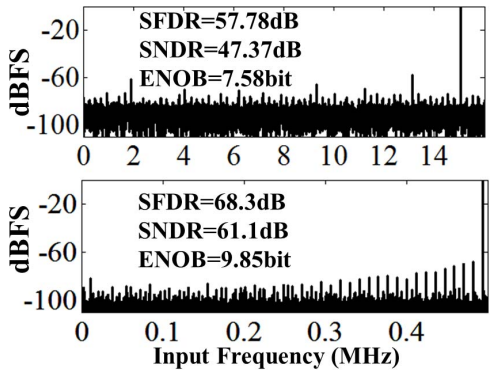


Fig. 14. Measured spectrum of the two ADCs with a near-Nyquist tone.

TABLE V
ADC DESIGN TIME OF THE PROPOSED FLOW

Manual Design time	Front-end (inc. optimization)	Back-end (inc. optimization)
SAR logic	days → minutes	days → minutes
DAC	minutes → seconds	days → seconds
S&H	hours → minutes	hours → seconds
Comparator*	days → seconds	hours →seconds
ADC top	minutes → seconds	days → seconds

*** Design time for the library is days**

The automated design flow successfully reduces the total ADC design time down to minutes level (Table V), after initial construction of the libraries, templates, and code. In particular, both the front-end and back-end design times of the SAR logic are reduced significantly from days to minutes. The time to prepare the digital design flow is at minutes level. For the DAC, the design time is reduced down to minutes level thanks to the automated layout generation. For the S&H circuit, it takes around three iterations for the front-end design through the LUT and fine tuning, and the layout can be generated automatically. The layout template preparation time for the DAC and the S&H is in few hours. The comparator library preparation time is still in days since it needs to be manually designed but this has to be done only once. The ADC top integration is reduced from days to seconds. Overall, thanks to the automated generation of all the subcircuits, the design time needed for one iteration is greatly reduced. In this way, it speeds up the design time considerably, even for a designer with less expertise. At the same time, the performance of

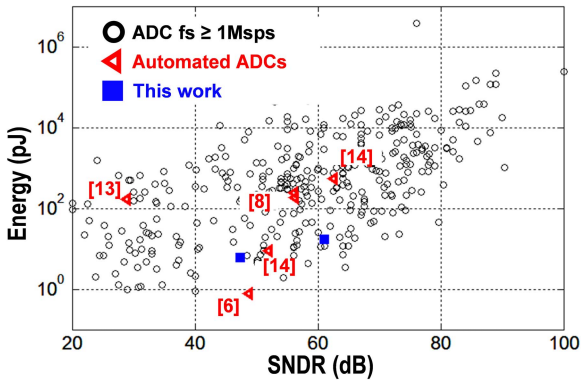


Fig. 15. Benchmark with the state of the art.

TABLE VI
PERFORMANCE SUMMARY OF THE TWO ADCS

	[13]	[14]				[6]	This work			
Experiment	Meas.	Meas.	Sim.	Meas.	Sim.	Meas.	Meas.	Sim.	Meas.	Sim.
Automation level	Schematic Layout	Schematic				Layout	Schematic Layout			
Architecture	Stochastic	SAR				SAR	SAR			
Technology (nm)	90	180		90		28	40			
Resolution (bit)	-	12		10		9	8		12	
Speed (MSps)	210	0.6		50		20	32	32	1	1
Supply voltage (V)	1.2	3.3		1		0.69	1	1	1	1
INL (LSB)	-	2.79	0.1	1.41	0.06	-	0.7	-	4.3	-
DNL (LSB)	-	5.24	0.22	2.81	0.1	-	0.97	-	1.55	-
Nyquist SNDR (dB)	29	62.7	73.9	52.1	61.9	48.84	47.37	49.3	61.1	71.6
Power (uW)	34800	328	53.5	440	155	15.87	187	192	16.7	16.8
FoM (fJ/c.s.)	7943	500	22	26.7	3.1	3.51	30.7	25.1	18.1	5.4
Meas. vs. Sim. ratio	-	22.7		8.6		-	1.2		3.4	

the ADC is maintained (Fig. 15), showing a good balance between the design time and ADC performance compared to the state of the art [23]. Compared to other synthesized ADC approaches (Table VI), this paper is one of the few to automate both schematic and layout design while it achieves good power efficiencies and the best matching between the predicted performance and measured results.

At present, the proposed tool aims for design assistance of SAR ADCs at medium speed and resolution. In the future, capacitor-mismatch calibration as in [21] can be integrated so that higher resolution ADCs can be generated. For high-speed ADCs, probably more attention is required to obtain a high-speed layout, the logic may need to be custom designed, and other circuit topologies might be required. Beyond SAR ADCs, the hybrid automation method could also be applied to other ADC architectures but it is most useful to relatively digital-intensive architectures. The most time-consuming part would be to build the layout templates and the component library where applicable. The more complex analog blocks are required, the less likely it is to benefit from the proposed tool.

VI. CONCLUSION

In this paper, a circuit design-driven tool with a hybrid automation approach for SAR ADCs is proposed. Compared to the prior art, instead of simplifying or modifying the circuit architecture to enable synthesis, our hybrid performance-selective approach allows the employment of appropriate circuits for performance and automation for design

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time reduction. In this way, the design time is reduced without compromising the performance, meanwhile still keeping the possibility to control performance-critical devices through programming, enabling a balance between automation and customization. The implemented prototypes achieve a performance that is sufficiently competitive with full manual designs while strongly reducing the design time and having an accurate control over the performance goal.

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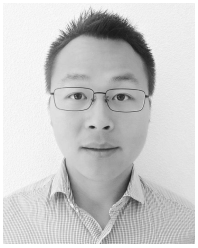
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